Natalie Hall

AI Applied Engineer

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WHY YOU SHOULD HIRE ME

- **Building Practical AI Systems:** You need someone who can turn AI from a concept into something usable. I analyze teams' needs & their processes to design & test workflows that structure prompts, reasoning, & outputs into repeatable systems aligned with real-world use.
- **Bridge Users & AI:** Many AI tools fail at usability. My background in UX & development allows me to build human-centered AI solutions that people can actually use effectively.
- **Ship & Maintain Real Systems:** I've worked on & maintained large-scale production environments (625+ websites), so I understand reliability, performance, & production constraints.
- **Rapidly Experiment & Iterate:** AI changes fast. I actively prototype, test, & refine workflows to quickly identify what works & what doesn't.

AI SYSTEMS & AUTOMATION EXPERIENCE

Jan 2025 – Present

Independent AI Systems Developer

ChatGPT, Claude API, Python, Async Programming, CLI Tooling

- Built a structured AI agent development framework using plan-based execution & iterative, deliverable-driven development, with deterministic validation through exact command-output matching to replace trial-&-error workflows with structured plan → execute → verify cycles, ensuring correctness across multi-step workflows for reliable & repeatable system generation.
- Built an AI-powered content aggregation pipeline that ingests newsletter data, performs embedding-based clustering, & generates structured daily digests using LLMs
- Designed a multi-stage AI processing system including parsing, semantic deduplication, AI-based filtering, & batched generation to handle LLM token constraints
- Implemented embedding-based similarity via Sentence Transformers to group similar content & reduce duplication across sources
- Developed structured prompting workflows using JSON schema outputs & tool-based interactions with LLM APIs
- Built resilient pipeline behavior with logging, validation checks, & fail-safe handling for partial or inconsistent outputs
- Built & deployed a production-ready AI chatbot on Hugging Face Spaces using Python & OpenAI APIs, incorporating a lightweight RAG approach to provide context-aware responses from personal data sources
- Continuously prototype & evaluate emerging AI techniques to improve system performance & workflow design

SKILLS

AI Systems & LLM Engineering: Claude API (Anthropic), OpenAI API, Gemini, DeepSeek, Ollama, Groq (Llama), LLM Pipelines, Prompt Engineering, Tool Use, RAG, MCP

Programming & Engineering: Python, JavaScript, Async/Await, CLI Tooling/Scripting, APIs (REST), Git/GitHub, HTML, CSS

NLP & Embeddings: Sentence Transformers, Text Embeddings, Semantic Similarity, Clustering

Systems & Architecture: Multi-stage Data Pipelines, Token-Aware Batching, Workflow Design, Agent Frameworks, Deterministic Validation

Tools & Frameworks: Cursor, Hugging Face, Gradio

Human-Centered AI, Systems Design & Problem Solving: User-centered system design, translate ambiguous needs into structured solutions, behavioral analysis & user insight, workflow optimization, information architecture, cross-functional collaboration, rapid iteration

CORE ENGINEERING EXPERIENCE

April 2022 – Nov 2025

Sesame Communications – Contract Web Developer

HTML, CSS, JavaScript, PHP, jQuery, Photoshop, Salesforce & Responsive Design

- Edited & improved 625+ dental websites by applying UX principles, accessibility standards, & research-informed enhancements
- Conducted informal user discovery to identify pain points & collaborate cross-functionally to align design decisions with user goals
- Performed QA & cross-browser testing to ensure consistent, inclusive experiences across devices
- Regularly optimized assets for speed & performance to support more reliable, accessible service delivery

Notable Achievements:

- Completed onboarding & training in just 4 days (quarter of the estimated time)
- Mastered my role quickly, which led to being entrusted with multiple tasks outside my initial scope to support team efficiency & project needs
- Ensured 100% accuracy when uploading files, avoiding code overwrites or rollback requests despite working without version control

July 2023 – July 2025

HostingCT – Contract Web Developer & Product Manager

WordPress, HTML, CSS, JavaScript, Photoshop, AI, CMS, Microsoft Office, Python

- Designed & updated websites for small businesses, prioritizing ease of use & client goals
- Led UX initiatives including user flows, content strategy, & usability testing to improve engagement & simplify site experiences
- Collaborated with cross-functional teams to align product decisions with user needs & technical feasibility
- Created email automation tools for outreach & retention using Python

Notable Achievement: • Built a retention email tool that increased client engagement by 20%.

June 2017 – October 2019

Bitlathe – Web Developer

JavaScript, jQuery, HTML, CSS, Git, Linux Command Line, JSON, Bootstrap, React, Vue, Grunt

- Developed internal & client-facing tools with a focus on stakeholder alignment & real-world users
- Maintained sites with version control & command-line tools for efficient updates & debugging
- Worked closely with senior developer & mentor to improve systems & design processes

May 2012 – March 2017

HDH Associates, P.C. – Marketing Director & Web Developer

HTML, CSS, Photoshop, Dreamweaver, Microsoft Office, SEO

- Led the redesign & maintenance of 3 corporate websites, focusing on clarity, modern design, & usability
- Managed all digital marketing assets & proposals, aligning design with organizational goals
- Transitioned the company from WYSIWYG editors to custom-coded solutions for improved flexibility & performance

INDEPENDENT DEVELOPMENT, CONSULTING, & RESEARCH

October 2020 – January 2022

The Undead Institute – Software Development Consultant

JavaScript, HTML, & CSS

- Provided usability feedback & reviewed code examples for a published book series on JavaScript, HTML, & CSS
- Edited technical documentation & helped refine examples for readers of varying skill levels

July 2021 – October 2021

iOS Application Developer (Bootcamp Program)

Swift, Xcode, UIKit, Firebase, CocoaPods, Objective-C

- Led the planning & development of multiple iOS apps, including group projects with a focus on design & user experience.
- Combined technical learning with user empathy & real-world testing

March 2004 – Present

Independent Web Developer & Digital Media Producer

HTML, CSS, PHP, WordPress, Dreamweaver, Final Cut Pro, Adobe Creative Suite, Microsoft Office

- Design, build, & launch websites, portfolios, & apps tailored to a wide range of user needs & technical literacy levels
- Apply UX principles to deliver responsive, branded designs focused on usability, accessibility, & content clarity
- Create video, visual, & written content to support user understanding, engagement, & business goals

EDUCATION

2025 – Present

Online Self-Directed Learning for Agentic AI

- Dynamous AI Mastery: Agentic Coding Course & AI Agent Mastery Course
- Udemy: AI Engineer Agentic Track: The Complete Agent & MCP Course

2015 – 2024

Online Self-Directed Learning for Web & Software Development

- Udemy • Udacity • Treehouse • Wes Bos • Codecademy • W3Schools

2021

iOS Development

- Devmountain (Bootcamp) • Hacking With Swift

2004 – 2008

BS in Media Studies: Production Technology

- Radford University, Radford, VA